

Maciek Czarnik

iOS Engineer

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Summary

For thirteen years, I've been successfully delivering value as a professional iOS Engineer and Leader. In my career the ability to read and understand code, sense and avoid the complexity, learn quickly, understand what does good design mean as well as empathy and great communication skills were always my advantages as a team member. I spent a part of my career working in a startup environment, where things are changing fast, and the most important thing is to deliver working solutions quickly, but passion to craft beautiful, well built, reliable, extendable and maintainable solutions lays deep in my heart and is always the ultimate goal. Until now I have been working on over 50 iOS and macOS apps including delivering value at every possible stage of the development.

Employment History / Work Experience

Fortunity Labs – Owner, Engineer, Designer (January 2023 – Present)

I am working on a new app dedicated to creating kinetic typography effects on videos, and investing my time in learning Metal and Graphics Engineering in general.

Earlier this year, I created and released the RPLY app. It's a document-based SwiftUI app built from the ground up with The Composable Architecture and Test Driven Development. I took the responsibility for the entire app (concept, design, and development). The app is still under development (aiming to make it feature complete and apply some nice design). Feel free to check it in the [App Store](#) or read the [release note](#) highlighting my way of thinking about this product development.

RDX Works – Senior iOS Engineer, Remote Contractor (June 2023 – October 2023)

In a team of four engineers we were developing an innovative crypto wallet built on top of our custom technology. The app is built with 100% SwiftUI, The Composable Architecture leveraging a modern Swift features such as structured concurrency (aka `async await`) and extensive tests of the logic. My main focus was to help the team to deliver as many features as possible for the [1.0 release](#).

Linearity – UI Engineer (January 2021 – December 2021), Lead UI Engineer (January 2022 – December 2022)

Initially, I got hired as a UI Engineer - a specialist responsible mainly for the development of the iOS and macOS user interface. Still, quickly I got my hands dirty with the rest of the system. Alongside 20-30 engineers in the company and 5-10 engineers in my team, I'm working on Vectornator - one of the top graphic design apps in the App Store. One of my early successes is the delivery of Quick Actions in less than 3 months. My next big assignment was the Color Widget and the Color Picker improvements. I introduced techniques such as Playground Driven Development and SwiftUI previews to iterate over decoupled UI modules fast (mostly UIKit). When I got promoted to a leadership position I focused more on team-level coding and process improvements, and the architecture of the system. My main accomplishment in that area is the delivery of architectural decision records that support multiple business goals and ensure truly Agile and collaborative work. I have been responsible for facilitating the adoption of the new approach, and coaching team members in that area.

We've been featured in the App Store and Apple events multiple times. This year we've been listed as a finalist for the Apple Design Award in the Interaction Design category.

Packs – Co-Founder (July 2020 – December 2020)

With two friends (one Graphics Engineer and one Domain Expert) we were working on an iPad and macOS app dedicated to visualisation of flat designs on 3D packaging models. I've been responsible for the native side of the project using SwiftUI. I was also dedicating my time to get up to speed with the most recent SDKs and features of the Apple ecosystem (mostly SwiftUI and Combine), studying the SwiftUI Lab blog, Composable Architecture by Point-Free and Thinking in SwiftUI book by objc.io.

Sweet Studio – Remote iOS Engineer (September 2019 – June 2020)

I was working on Strucc app – the creative tool for filmmakers. My role in a distributed team of three people was focused around user interface and integration of underlying services and the rendering engine. My main achievements were enabling continuous deployment and establishing the clear architectural borders between the user interface, underlying business logic, rendering engine, and services. These facilities contributed significantly to the velocity of the development and were crucial in pivoting from version 1.1.0 to 2.1.0. I've been also responsible for the implementation of a series of experiments for validated learning.

Unemployed (May 2019 – August 2019)

After leaving Showpad, I decided to dedicate my time to education and another business trial. The education spanned across Functional Programming, GraphQL, AWS services, SwiftUI, Combine and Recommender Systems. As a business trial, I've been trying to bring up my software house focused on the development of user interfaces.

Showpad – Remote Senior iOS Software Engineer (November 2018 – April 2019)

I have been working in a remote team spread across the globe (two people in Poland, three in

Belgium and two in the United States). As the company acquired an e-learning platform (LearnCore) last year, we're working to bring it to the existing Showpad app. The expectations for the quality were pretty high, and my main focus was to implement and maintain the UI and controller layers for two main experiences (screens). I have introduced and fully leveraged the Playground Driven Development technique. I've been also responsible for bug fixing and maintaining the legacy part of the code base in both Swift and Objective-C.

Interact Software – iOS Developer (June 2018 – August 2018), Remote iOS Developer (September 2018 – October 2018)

In a cross-functional team, we're working on an enterprise-level intranet solution – an internal Facebook or LinkedIn kind-of app. When I joined the company the code and architecture were so overcomplicated that the development almost stagnated without delivering any significant end-user value. I did manage to sense that problem in one day, propose a solution and convince the whole development team (Android included) to pivot with the beginning of the next sprint. With the applied changes we did manage to release first version of the product in time, scope and budget. We have been working in Scrum framework with flat team structure (no platform leads or seniors). Except for coding and overseeing the other iOS developer by code review I've been managing our automation in Bitrise.

Stanwood GmbH – Remote iOS Developer (October 2017 – June 2018)

I have been working on a several projects including B2C and B2B apps for iOS, tvOS and watchOS platforms as well as an internal framework for a chat UI component. I have fully implemented a first version of the ON Air tvOS app. Except that my main duties were to maintain and implement new features in such projects as ePaper (multi-target magazine client app) and Calorie Tracker (personal weight control assistant). I have also been doing code reviews and presentations on topics such as clean code and architecture. We've been working in a very loosely defined methodology in a cross-functional teams with two iOS developers per team, while using Atlassian, Github, Bitrise and Firebase tools just to name a few.

F. Hoffmann - La Roche Ltd – Remote Lead iOS Developer (December 2016 – October 2017)

I have been leading dynamic 5-10 developers team spread across the country. We're rewriting an enterprise CRM client app written in Objective-C into Swift with focus on improving overall architecture, maintainability and performance of the previous solution. App uses Core Data as a database framework and communicates with Salesforce backend through the REST/SOAP API. My main duty was to assure a technical excellence of the solution by developing solution architecture, eliminating complexity, inspiring, guiding and overseeing developers by code review. We were using Scrum methodology on Atlassian Stack with Stash, Bamboo and JIRA.

Unemployed (June 2016 – November 2016)

After Contentment went out of business I focused on WWDC16 and education again. I have been focused on Swift 3, iOS 10 and the basics again – code quality, design patterns, SOLID principles, modern architectures, tools, Continuous Integration and Test Driven Development. As a side effect I've been building a client app for the Hypothes.is annotation tool using RxSwift and Moya and coding in the Open Source.

Contentment – Remote iOS Developer (November 2015 – April 2016)

Onsite iOS Developer (September 2013 – October 2015)

Contentment was the second company in the publishing industry that I've been working for. In the cross-functional (designers, front-end, back-end developers), multinational development team we have been using SCRUM with Pivotal Tracker, Trello and JIRA as a tools.

I have been fully responsible for iOS development in the company. I have been overseeing and developing a hybrid architecture of Contentment apps (client apps of the Padify and Puggig CMSs) based on Cordova for both iOS and Android. I have worked in both Xcode and Eclipse switching from Objective-C to Java and Swift 2 later on. I started with an existing codebase and quickly discovered (alongside with a CTO, who have joined the company soon after myself) the weaknesses of the system and during two years in Contentment we have gradually (in 3 iterations) changed and stabilized the structure of the system to make building process of the apps almost automated. I have designed and developed the Padify Reader SDK (uses Mantle, AFNetworking, CocoaLumberjack, Appledoc) which enables developers to easily embed Padify CMS content in any iOS app. My main duties were developing the mentioned above internal architecture and tools for our apps, preparing and distributing App Store and Ad Hoc builds (Fabric.io, Crashlytics), implementing new Cordova plugins (iBeacons, custom Push Notification providers like Urban Airship, Parse, Pushapps, Mixpanel etc, extending and fixing existing plugins, In-App Purchases with subscriptions and validation of the receipts) and working with Production Team to teach them how to build the apps and test their content in them.

We have successfully released over 30 iOS apps, but unfortunately Contentment is currently out of business, and most of the apps that we have released with them are now updated by other developers (these were magazine apps, so clients needed to switch to a different app provider to continuously release new issues). The clients included: Time Inc., ACCA Global, Motorsport Magazine, NOTION Magazine and many others.

Super Smart Apps – Owner (November 2012 – August 2013)

I decided to leave my previous company and dedicated myself to learning new things and trying to make money as a sole trader. During that period I acquired loads of knowledge about modern tools and workflows related to iOS development, as well as a lot of knowledge about e-business topics like e-marketing, lean thinking, business modeling tools (Business Model Canvas, Lean Canvas) and strategies of bringing product into market. I have been working on a couple of proof of concept apps in a wide range of domains (retail search engine and special offers app, real estate app with AR, Map Kit and Core Location, educational tool with SuperMemo algorithm) and released two apps (one for iOS and another for Android):

- Ideapp – Save Your Ideas!

It's my first iOS app released and fully implemented by myself. It featured modern (iOS 6 times), highly customized design prepared by my friend Radek Skrzypczak with interactive animations and custom transitions (using ADTransitionController – they were not yet an element of the UI Kit). In the project I used Git, CocoaPods, Google Analytics, Core Data, KVO, DCIntrospect and Reactive Cocoa. We decided to release it for free, but reviews in the App Store and fact that thousands of people still use it around the World are highly rewarding.

- [Dowcipy P.Rosiak](#)

It's my first Android app made for Zielona Sowa publishing house released and fully implemented by myself. It's a very simple app written in Java with JSON database, which simply allows to browse jokes.

Implere Ltd, lately rebranded to Connected Digital – Senior Developer (August 2011 – October 2012)

As a Senior Developer, with two other developers (cross functional team), I was responsible for ReaderLib – the mobile client of ZineGenie CMS (the flag product of the Implere company). ReaderLib is a shared library for a number of apps, it is an electronic publications browser which allows users to buy, subscribe, browse, share and comment on articles and publications prepared in the CMS. Most of the time, I was working on the iOS platform, but I have also spent about 5 months working on the Android platform.

There were three main areas of the app: text view, page view and downloading/caching mechanism. CMS served the content in XML files, which were downloaded, cached and reparsed to adapt to the actual device specifications. The text view were presenting content in the HTML form (from reparsed XML) and page view allows users to browse magazines in image mode. The architecture and a working app was established before I joined the company, so my main duties were designing, implementing and improving functionalities for new layouts and views, turning images into PDFs in a page mode, adding audio/video clip playback support, Push Notifications for particular clients, In-App Purchases, custom adverts, Google Analytics tracking, image galleries, as well as embedding new elements on page and text mode views. Taking care of bug fixes, performance and stability issues.

During my work at Implere I worked on a number of applications, some of which are detailed below (there were few other ad-hoc and yet unreleased applications):

- [Travel Africa Magazines](#)
- [Physics World](#)
- [Chemistry World](#)
- [Page Lizard - DigitalPC Digital Editions](#)
- [Dive Magazine \(iOS, Android\)](#)
- [Which Right Choice \(Android – Not available in the Google Play any more\)](#)
- [CRN UK Magazine \(Not available in the Google Play any more\)](#)

PGS Software S.A. – iOS Developer, .NET Developer (July 2010 – July 2011)

After graduating from the University I gained my first iOS Developer experience here. I didn't look for a job, as I've been trying to turn my masters project into business, but one day I received an offer with a role in a newly created mobile development team. I decided to jump in and acquire a new skill related to my main professional interest. As a Junior iOS Developer I've been part of iOS Development team made of one Senior and another Junior developer. We've been using SCRUM methodology (using JIRA) and working in Xcode with Objective-C (SVN as CVS) with iOS versions 3 and 4 producing native iPhone/iPad/iPod Touch as well as universal apps. I've taken part in 4 iOS projects:

- [A2Dominion New Homes \(Not available in the App Store any more\)](#)

The app was a real estate agency client application which enables users to browse current offers and get the basic info like contact numbers, photos and location of the houses for sale or rent on the map and in a search view. I have joined the company in the middle of the project, so my first tasks were about chasing some UI issues (not in Interface Builder yet) and memory problems. Later with a bit of the Senior support I've implemented a gallery module for the app. My main focus was on getting to know the basics: Objective-C, Foundation, UI Kit and Cocoa Touch architectural patterns.

- IMAGEM Creative Services (Not available in the App Store any more)

It was something like limited Spotify app. After successful login you've been able to browse and stream mp3 music through your iPhone/iPad device. I've been partly designing and implementing a RESTful client-server communication (JSON) along with a data model classes and underlying SQLite database. My good understanding of audio and passion for music resulted in fact that I've been entirely responsible for the multi-threaded (buffering of the next track in the playlist on a second thread) Core Audio streaming engine. Thanks to my patience and dedication to solve problems I've been given a task to ensure a stability and memory usage efficiency of the app, which was quite challenging in pre-ARC times.

- Apps for ROSINANTE&CO (1, 2, 3, 4, 5)

These are the book-like apps, that simply consisted of high-quality images, text and audio. They were all based on a multi-target project. I was responsible for the correct (pixel perfect) positioning and rendering of the text using Core Text framework in many different languages (Internationalization support). I have made a simple OS X tool to help with getting the text positions on particular slides into backing .plist files. I've also made a browser UI view for other ROSINANTE&CO apps available on the App Store by using iTunes Connect API.

- Music toplists

It's a social app, which is basically about sharing favorite music tracks (Youtube, Last.fm). As a team we have implemented Push Notifications, designed and implemented the full CRUD RESTful communication with the server. I've implemented most of the skeuomorphic designs (strongly supported by Core Animation) as well as Facebook and Twitter integration for sharing a top lists.

I have taken part in one more project in .NET technology. It was a project for LM Wind Power company and with two other developers we produced a software which main goal was to assist the operator in making a decision if the tested wind blade was durable and qualifies to use or it should be taken down and destroyed (based on the computer vision analysis). I was responsible for cracking the internal structure of a data (USG images) in a binary input files. I have also entirely designed and implemented the AI module that makes calculations on the data and provides related measurements and a final suggestion on what to do with a tested blade.

Things That I Believe In as a Professional

- The most important code is the interface. Suboptimal decisions in the single line of the interface can influence tens, hundreds, or even thousands of lines of code that depend on it. On the other hand, good, essential interface is an invaluable communication and collaboration medium.
- The most valuable processes are the ones that result in tightening the Build, Measure, Learn loop. Also, in the context of coding.
- Software architecture is about universalities, boundaries, degrees of freedom, and granularity that support technical excellence under strict, strategical business requirements.
- Wrong framing of the problem leads to inferior solution. “Nobody owns the problem. Every problem is universal”, after Russell Ackoff.
- If you want to go fast, go alone, if you want to go far, go together.
- Most principles of good design are universal and applicable to the software development too. After Massimo Vignelli - “If you can design one thing, you can design everything.”
- Adopting single source of truth and unidirectional data flow eliminates most confusion out of existence.
- Evolutionary design trumps big design up front.
- “Make it work, make it right, make it fast”, after Kent Beck.
- Complexity should be kept as private as possible.
- Consistency over innovation.
- People over process.
- Ego is your enemy.

Education

October 2004 – June 2009

Rzeszów University of Technology, the Faculty of Electrical and Computer Engineering

Master of Sciences, Engineer in Information Systems

Master Thesis – “Software engineering on Symbian S60 Platform”

The 10-term course included wide spectrum of computer science topics such software engineering, databases, operating systems, networks, artificial intelligence, signal theory and applications, electronics and more IT-and-engineering-related subjects. I have been awarded the scholarship for each year of studies. I managed to achieve the highest note at the overall classification of the qualification.

September 2000 – June 2004

Comprehensive High School no. 1 in Rzeszów

Class profile: Mathematics and Information Technology

Interests / Activities / Skills

Foreign Languages

I have good knowledge of verbal and written english.

Outside the Office

I'm interested in design (in a broad sense) and the knowledge of good design principles inspires my coding and architectural decisions every single day. My hobby is music. I have finished Primary Music School on the piano, the State School of Music on the double bass and I have started a music course on jazz composition and arranging on the Music Academy (currently suspended). I am a fan of bike riding, TRX and healthy lifestyle I am keen on e-business, art and new technologies.

References

My references are available on my [LinkedIn profile](#).

You can also check my profile on [Stackoverflow](#), [GitHub](#), [Twitter](#) and my blog at [Medium](#).